Sagalpreet Singh

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sagalpreet *sagalpreet* *Google Scholar*

Education

2026 – 2029 (tentative) **University of Oxford**, Oxford, United Kingdom
DPhil in Computer Science, OATML Group
Supervisor: Prof. Yarin Gal; funded by LASR

2019 – 2023 **Indian Institute of Technology Ropar (IIT Ropar)**, Punjab, India
9.31/10.0 Bachelor of Technology, Computer Science and Engineering
10.0/10.0 Concentration in Artificial Intelligence

Work Experience

Aug'26 – Oct'26 **Proximal**, *Member of Technical Staff*, Bengaluru, Karnataka, India.
Focus: Reinforcement Learning; Long-Horizon Agentic Environments

› Built **long-horizon RL environments** for training and evaluating agents on multi-step, verifiable tasks

Aug'24 – Jul'26 **Google DeepMind**, *Pre-Doctoral Researcher*, Bengaluru, Karnataka, India.
Focus: Reinforcement Learning; Continual Learning; Domain Adaptation
Advisors: Dr. Rishi Saket, Dr. Aravindan Raghuveer

› Designed RL algorithm with **convergence guarantee** and **sample complexity** for preventing mode collapse in multi-goal environments; conducted systematic training runs and checkpoint evaluations on multi-step coding environments; patent filed; presented demo at NeurIPS'25

› Engineered scalable, fully-verifiable **RL-driven curriculum generation** engine for training **Gemini Thinking** models; designed verifiable reward functions and training signal pipelines; contributed 38% weighted data; >10% improvement on Math and NL Reasoning

› Investigated **reward hacking** phenomena and conducted trajectory-level analysis in multi-step RL settings; systematic evaluations across training checkpoints to characterize emergent failure modes

› Proposed novel objective for **domain adaptation** to weakly supervised target domain with label proportions; derived **generalization bounds**; **SOTA** on public benchmarks; multi-million dollar impact on Google Ads

› Designed **continual learning** system via active retrieval in LLM-based agents for persistent non-parametric memory and compounded lifelong improvement

Jun'23 – Aug'24 **Oracle AI**, *Member of Technical Staff*, Bengaluru, Karnataka, India.
Focus: Text-to-Speech; Transformers; Distributed Training; Backend Engineering
Manager: Phanindra Mankale

Jan'23 – Apr'23 **LASII Lab**, *Undergraduate Researcher*, IIT Ropar, Punjab, India.
Focus: Style Transfer, Diffusion Models, Audio Processing
Advisors: Prof. Abhinav Dhall

Mar'22 – Oct'22 **GTML Lab**, *Undergraduate Researcher*, IIT Ropar, Punjab, India.
Focus: Machine Learning, Human-AI Teaming, Bayesian Inference
Advisors: Prof. Shweta Jain, Prof. Shashi Shekhar Jha

Jun'22 – Jul'22 **Oracle Integrations Cloud**, *Software Engineering Intern*, Remote.
Focus: Backend Engineering
Manager: Muthu Palanisamy

Jan'22 – May'22 **IPSA Lab**, *Department Engineering Project*, IIT Ropar, Punjab, India.
Focus: Android Development, Backend Engineering
Supervisor: Prof. Puneet Goyal

Publications

RLC 2026 (Oral) **Sagalpreet Singh**, Rishi Saket, Aravindan Raghuveer. 2026. Dense and Diverse Goal Coverage in Reinforcement Learning.

UAI 2025 **Sagalpreet Singh**, Navodita Sharma, Shreyas Havaldar, Rishi Saket, Aravindan Raghuveer. 2025. Learning from Label Proportions and Covariate-shifted Instances. Proceedings of the Forty-first Conference on Uncertainty in Artificial Intelligence, PMLR 286:3920-3938, 2025.

AAMAS 2023 (Oral) **Sagalpreet Singh**, Shweta Jain, Shashi Shekhar Jha. 2023. On subset selection of multiple humans to improve human-AI team accuracy. In Proceedings of the 2023 International Conference on Autonomous Agents and Multiagent Systems (AAMAS '23). International Foundation for Autonomous Agents and Multiagent Systems, Richland, SC, 317–325.

Patents

2025 **Sagalpreet Singh**, Rishi Saket, Aravindan Raghuveer. Dense and Diverse Goal Coverage in Reinforcement Learning. (Filed)

Talks and Demos

Aug 2026 **Dense and Diverse Goal Coverage in Reinforcement Learning**, *RLC 2026 (Oral)*, Montreal.

Dec 2025 **Dense and Diverse Goal Coverage in Reinforcement Learning**, *Google DeepMind Booth*, NeurIPS 2025, San Diego.

Dec 2024 **Learning from Label Proportions and Covariate-shifted Instances**, *Chaupal, Google DeepMind*, Bengaluru.

Jun 2023 **On subset selection of multiple humans to improve human-AI team accuracy**, *AAMAS 2023 (Oral)*, London.

Professional Projects

Google DeepMind

2025 **Verifiable Problem Generation for Gemini Thinking**

› Engineered scalable, fully-verifiable **RL-driven curriculum generation** engine for training Gemini Thinking models; designed verifiable reward functions and training signal pipelines for mathematical and logical reasoning tasks.

› Contributed 38% weighted data in the Gemini thinking data revision leading to >10% improvement on downstream tasks like Math and NL Reasoning.

2026 **Long-Horizon RL for Filesystem Exploration and Navigation**

› Built long-horizon RL training data for filesystem exploration and navigation in Gemini models across multiple scales.

› Achieved >6 pp improvement on downstream agentic evals for XS model with headroom confirmed for larger models.

› In collaboration with Code Strike and Recursive Self-Improvement teams.

2026 **Active Retrieval for Continual Learning in Agents**

› Enabling continual learning via active retrieval in LLM-based agents to have persistent non-parametric memory that leads to compounded and lifelong improvement through experience.

2025 **Hybrid LLP for Ad Conversion Modeling**

› Proposed novel objective function for domain adaptation to weakly supervised target domain consisting of aggregated data with label proportions.

› Multi-million dollar impact on Google Ads.

Oracle AI

2024 **Text-to-Speech Model Development, Training and Deployment**

› Built Oracle's transformer-based in-house Text-to-Speech model supporting few-shot voice cloning.

› Worked with vendors for high fidelity data procurement.

› Trained model on distributed Slurm cluster and ran extensive evaluations against baselines showing SOTA WER/CER scores.

› Deployed the model via Triton inference server, created API spec, implemented API backend, and generated SDK and documentation.

› Handled customer calls for integrating the service into their applications.

2023 **Supporting Async Job Deletion in ASR Service**

› Added support for deleting async transcription jobs with rigorous checks in place and implemented corresponding UI changes.

Oracle Integrations Cloud

2022 **Support customer-tenancy based quota for email notifications**

› Developed circuit breaker to handle the noisy neighbor problem in the email notification microservice that is used more than a million times a day

› Modified UI to allow users to enter SMTP credentials.

› Was offered full time role at the end of the internship.

IPSA Lab (IIT Ropar)

2022 **SAMPAN App and Dashboard**

› Worked on SAMPAN Android App used by 1200+ Anganwadi (courtyard-shelter) workers in India to log 50000+ data points every month.

› Designed, developed and deployed dashboard on IIT Ropar servers for district officials.

› Featured in The Tribune.

Selected Open Sourced Projects

Google DeepMind

2026 **Dense and Diverse Goal Coverage in Reinforcement Learning**

› Designed RL algorithm with convergence guarantee and sample complexity for preventing mode collapse in multi-goal environments.

› Open sourced Jax based implementation of our algorithm, baseline algorithms as well as adaptation of Brax environments to multi-goal setup.

2025 **Learning from Label Proportions and Covariate-shifted Instances**

› Proposed novel objective function for domain adaptation from fully supervised source domain to weakly supervised target domain comprising aggregate data in form of label proportions.

› Derived generalization error bound and corresponding sample complexity.

› Achieved SOTA results on public datasets like Criteo Sponsored Search Conversion Log and IPUMS.

› Open sourced TensorFlow based implementation of our method as well as baseline techniques.

IIT Ropar

2023 **On Subset Selection of Multiple Humans to Improve Human-AI Team Accuracy**

› Developed an efficient algorithm based on Bayes' rule to combine multiple human labels with logits from ML models.

› Our method significantly outperformed naive combination methods in image classification on CIFAR-10H.

Honors and Awards

2023 **Google Research travel grant (USD 2000) for AAMAS 2023**, Google Research.

2023 **Microsoft Research travel grant (INR 120k) for AAMAS 2023**, Microsoft Research.

2023 **Best poster award**, *Research Scholar's Day*, IIT Ropar.

2019-23 **Institute Merit Scholarship for being in the top 7% of the batch**, *IIT Ropar*.

2022 **Selected for Amazon ML Summer School and offered Applied Scientist Internship**, Alexa AI.

2022 **Best B.Tech. project award**, *Technology Day*, IIT Ropar.

2019 **Top 0.1% Merit Certificate in Physics**, *Senior Secondary Exam*, CBSE.

2017 **NTSE Scholarship (given to top 1000 students in India)**, Government of India.

Competitions and Rankings

Competitive Programming	2022: **India Top 20**, *Oppo Inspiration Cup*, Oppo.
	2022: **Global Rank 238**, *Google Kick Start*, Google.
	2021: **Global Rank 198 (18th India)**, *Google Hash Code*, Google.
	2021: **Global Rank 84**, *Codegoda*, Agoda.
	2021: **Global Rank 83**, *Newton's Grand Contest*, Newton School.
	2021: **Rank 7**, *AMEXpert Smartathon*, American Express.
Olympiads	2019: **National Top 1%**, *National Standard Examination in Physics*, IAPT.
	2017: **State Top 1%**, *National Standard Examination in Junior Science*, IAPT.

Responsibilities

Reviewer	2025: ACL-ARR (May)
	2026: AISTATS, ICML, UAI, NeurIPS.
Teaching	2023: Deep Learning, Guvi.
	2022: Algorithms and Data Structures, Coding Club, IIT Ropar.
Senior Member	2023: Inter IIT Tech Contingent, IIT Ropar.
Representative	2022: Computer Science and Engineering (Batch of 2019-23), IIT Ropar.
Representative	2020: Coding Club, IIT Ropar.
Organizer	2020: Fury Road Event, Robotics Club, IIT Ropar.

References

Available on request.